

TOI-2046b

R-2327

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TOI2046_250802 · created 2026-08-18T22:52:20+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460889.984 +/- 0.02 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.159 +/- 0.0098
Transit depth (Rp/Rs)^2	2.53 +/- 0.31 [%]
Orbital Inclination (inc)	85.2 +/- 2.31
Airmass coefficient 1 (a1)	1.084 +/- 0.012
Airmass coefficient 2 (a2)	-0.056 +/- 0.078
Scatter in the residuals of the lightcurve fit is	1.07 %
Best Comparison Star	#2 - [54, 314]
Optimal Aperture	4.1
Optimal Annulus	10.55
Transit Duration (day)	0.057 +/- 0.02

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.159 $\pm$ 0.0098 vs 0.1213 (deviation 3.75 σ)
Duration fit vs published	0.057 d vs 0.1033 d (deviation 2.26 σ)
Mid-time O-C	42.1 min
Depth SNR	15.8
Residual scatter	1.07 %

Light curve

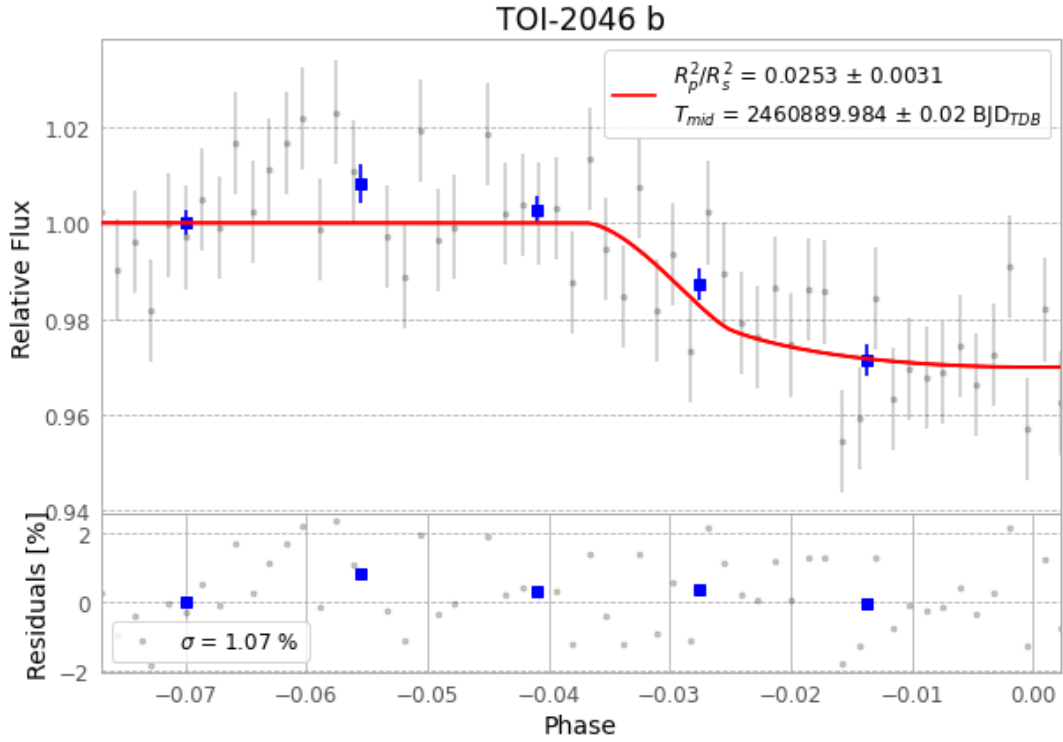


Figure 1 — FinalLightCurve_TOI-2046 b_2025-08-02.png (SHA-256 76bdfb74b3bbbedad...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [399, 226], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 273242abf9403684...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

57 FITS frames (37 MB), TOI2046250802085116.FITS → TOI2046250802114215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI2046-250802.log (d17b6c152777...), FinalLightCurve_TOI-2046 b_2025-08-02.png (76bdfb74b3bb...), FinalLightCurve_TOI-2046 b_2025-08-02.csv (a8a294f0166d...)

Compute resources

Wall time 71.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-18 22:52Z.